Maastricht University

Department of Advanced Computing Sciences

TODO: Thesis title

Bachelor Thesis

Author: Vojtěch Kosatík
Student ID: i6351976
Supervisor: Prof. Dr. Adriana Iamnitchi & Dr. Charis Kouzinopoulos

May 13, 2026

Contents

1 Introduction **1**

1.1 Relevancy and Problem Statement 1

1.2 Existing Research and Detection Methods 2

1.3 Research Question and Objectives 2

1.4 Scope and Limitations 3

2 Background **4**

2.1 Visual Embeddings 4

2.2 Approximate Nearest Neighbor Search 4

2.3 Homophily in Social Networks 5

2.4 The FediData Dataset 5

3 Methodology **6**

3.1 Dataset and Preprocessing 6

3.2 RQ1: Visual Homophily Test 8

3.2.1 Results for RQ1 TODO: for now here later move to results with proper discussion 9

3.3 RQ2: FAISS Benchmarking 9

3.4 RQ3: Coordination Detection 9

3.5 Ethical Considerations 10

4 Results **11**

4.1 Overview of Findings 11

4.2 Results for RQ1 11

5 Discussion **12**

5.1 Interpretation of Findings 12

5.2 Comparison with State of the art and current research 12

5.3 Limitations and Future Research 12

6 Conclusion and Future Work **13**

6.1 Conclusion 13

TODO: Abstract as last

Sample text for abstract here.

TODO: Keywords: ..., ..., ...

Chapter 1

Introduction

1.1 Relevancy and Problem Statement

In today's world, largely defined by social interactions and exchanges of opinions, the internet is used in more and more extensive ways of mediating such connections. A popular example of a medium that facilitates these interactions is social media platforms. While they can generally bridge physical distances where social interactions were dependent on the speed of medium used (e.g. telegraph or postcards) if not impossible entirely in the past.

In very recent history, there have been cases in which social media platforms were hijacked and abused by individuals or organizations with malicious intent toward the general public. Badawy et al. showed how the Russian Internet Research Agency ran a large-scale influence operation on the platform Twitter prior to the 2016 presidential election. During this operation, 221 accounts were active, socially divisive, and detected only retrospectively after damage was already done. While Twitter is an example of a deeply centralized social media platform, meaning that a single entity is in control of user data and moderating the content, there is an alternative to this in the form of decentralized platforms (e.g., Mastodon or Pixelfed), where anyone can create and host their own instance of the platform. A prime example of how these platforms can then communicate with each other is Fediverse, which is only connected via a common protocol each of them uses - ActivityPub. (TODO: Acitvity pub source) Mastodon being a part of the Fediverse.

These platforms are relevant in conversation regarding coordinated inauthentic behavior since it has also been documented during the 2024 US presidential elections that 47 coordinated Fediverse accounts were found posting pro-democratic messaging using URL and text similarity networks. So centralized and decentralized platforms alike are susceptible to these unwanted coordinated operations with Fediverse as a whole being heavily understudied and having a large amount of smaller communities under variying

management. The fact that platforms in Fediverse might behave differently and that it's under-studied FediData dataset was created which will be the primary dataset used in this study. (TODO: Cite FediData paper)

1.2 Existing Research and Detection Methods

While we know inauthentic behavior occurs even on platforms like Mastodon, it is also important to highlight existing detection methods for it. Although there are already methods of detecting individual social media accounts for automated behavior, such as BotRGCN, BECE, or SGBOT, which utilize features from the social graph constructed from follower to followee relationships and textual information from user content, this approach on its own falls short of coordinated behavior spanning across multiple accounts. On the other end of the spectrum, an example of where the state of the art currently is at detecting coordinated campaigns is Graham et al.'s Coordination Network Toolkit (CNT) and Luceri et al.'s framework and machine learning model aimed at detecting coordinated state-sponsored Information operations (IOs)(TODO: cite state-of-the-art paper and Luceri paper).

However, what both the methods of inspecting a singular account for automated behavior and the methods of detecting a coordinated behavior with ulterior motives both have in common is the modality they leverage. As was already mentioned, they either utilize social graph features or text-based information such as user posts, metadata of the posts, or profile descriptions. Since Huang et al. have proven that multi-modality leads to better representation of the original dataset, visual modality presents itself as blind spot in current research in coordination detection literature.

Lastly, when working with user data from social media platforms, the sheer scale of information to process could pose challenges. Hence, the scalability of the proposed solution should be examined. For instance, Azizi etl al. (SIGMOD paper 2025 i thinnk) benchmarked graph-based methods up to 1 billion vectors, however spefcific recall-latency behavior on social media images isn't analyzed.

1.3 Research Question and Objectives

Based on the problems mentioned earlier and using specific technologies outlined in the methodology section, the following research questions were formulated: (TODO: should I go a tiny bit in depth of what CLIP, FAISS and pHash is since I already cover it in background section? Should I reference the section here now?)

1. To what extent does visual content similarity predict social connections on FediData?

2. What is the optimal recall-latency trade-off across FAISS indexing configurations for approximating nearest neighbor search on FediData image CLIP embeddings?

3. Does CLIP-based visual similarity detect known bot accounts with higher precision than perceptual hash (pHash) matching when applied to socially disconnected pairs, and does it capture semantic coordination that exact-copy detection misses?

1.4 Scope and Limitations

It is important to mention that this work does not attempt to create an independent system that detects coordinated operation of inauthentic behavior. That falls outside the scope and plan of this thesis. Instead, it attempts to introduce already existing technologies on modality that haven't been explored to enough depth, especially given the origin of the images coming from decentralized platforms rather than the more popular centralized counterparts.

Chapter 2

Background

2.1 Visual Embeddings

CLIP maps raw images to a fixed-size 512-dimensional vector using an image encoder. Although CLIP can also encode text, for our work, only the image encoder was used. The resulting vectors are constructed such that images with similar semantics have larger cosine similarity between the vectors, and images with contrasting meaning (e.g., a photo of a protest vs. a photo of bread) are pushed further apart in the vector space. The fact that CLIP is able to cluster based on semantics of a picture, thanks to its training rather than pixel-level statistics of an image, it is much better at representing the similarity of images with similar meaning, closer in contrast with methods that only detect similar pixel arrangements (described further in 3.4 TODO: comment on pHash there). source clip paper Radford

Every image (765,019) from the FediData dataset was therefore passed through the encoder once and stored as a 512-dimensional vector.

2.2 Approximate Nearest Neighbor Search

The FediData dataset has 57,760 users while other social media platforms like Twitter has over 500 million users (TODO: source). To search through these embeddings exhaustively, computing cosine similarity between every vector scales poorly. Even if we made a mean of all image embeddings posted by a particular user, for our FediData, the pairwise comparisons needed for all users would exceed 1.66 billion operations. Although this approach would be exact, it would fall short in practical scenario where multiple queries would be necessary on large-scale data. An approach that can circumvent this issue is Approximate Nearest Neighbor (ANN) searching addresses this by indexing all the image embedding vectors into specific data structures that allow large traversals across the vector space. Trading small loss in recall for significant gains in query speed. (TODO: cite

Johnson paper here)

2.3 Homophily in Social Networks

2.4 The FediData Dataset

FediData (FD)

Chapter 3

Methodology

3.1 Dataset and Preprocessing

After downloading the 765,019 images, the FD creators constructed one large folder and put all of them into it. The mapping of what image belongs to what user and what post of that user can be done by thte naming convention of those images FD creators made. Every image name can be split into 3 parts. On an example imaage: a_b_c.jpg, a refers to a user index by which we can map every image to its user, b refers to the post index since particular user can have multiple posts and c refers to image index of that post. Default value of c is 0 but in some cases a post can have multiple images.

Since the whole FD was download onto an external SSD, every time the operating system had to index all of the images it was taking significant amount of time every instance the external disk was plugged in. Therefore based on the naming convention of the individual images mentioned earlier the original flat directory was reorganized into per-user subfolders based on the user index a. This simple edit to original FD sped up the OS indexing significantly and made it easier to work with FD day-to-day.

After this change the script for making the embeddings could be made. Since this work was performed on M-series mac device MPS was selected at the beginning using pytorch where we then load CLIP model openai/clip-vit-base-patch32. This specific version was selected as it's the most often cited variant in literature, it was the perfect balance between its performance given the limits of machine it ran on and lastly it's able to produce 512-dimensional vectors which are then L2-normalized. Exactly what program for ANN searching expects as input. We then make batch limit of 64 images to limit memory usage and after the resulting normalized vector is obtained we store it in float32 array as that's the format the ANN library used in this work expects input in.

TODO: explain how in the FD paper researchers labeled few accounts as social bots. It was done by the researches themself and cross-checked against randomness with high Cohen's kappa of value of 0.876 (todo: don't forget to cite FD paper)

TODO: will probably have to move the following into results section

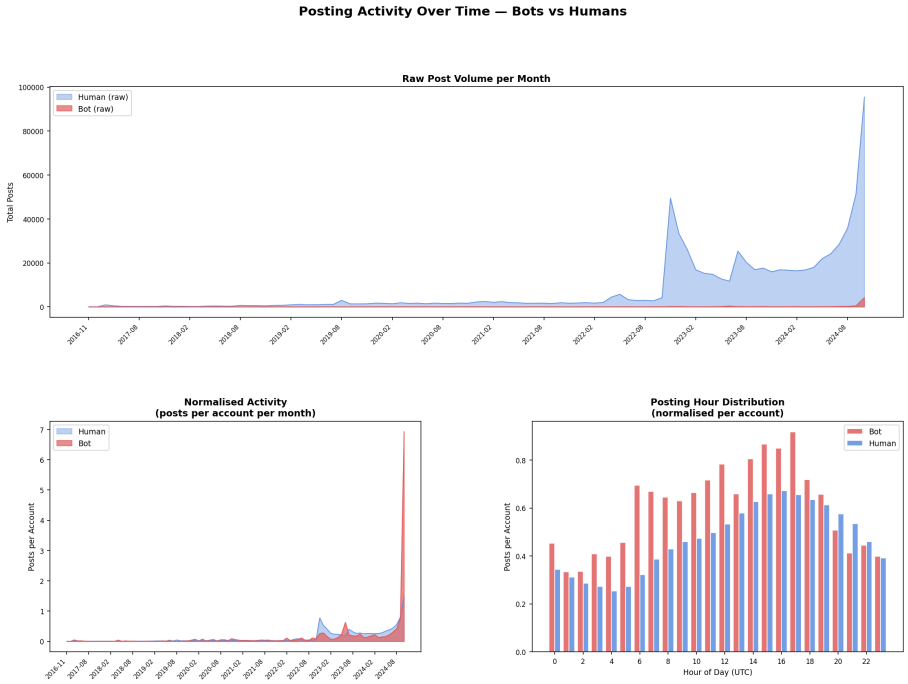


Figure 3.1: Showing account activity of FediData over time and hour of day

As shown in Figure 3.1, we can see an influx of raw post volume coincidentally with the upcoming US presidential elections in 2024 originating from social bots accounts. Additionally we can see that social bots outperform their human counterparts in posting much more regurarily during working hours.

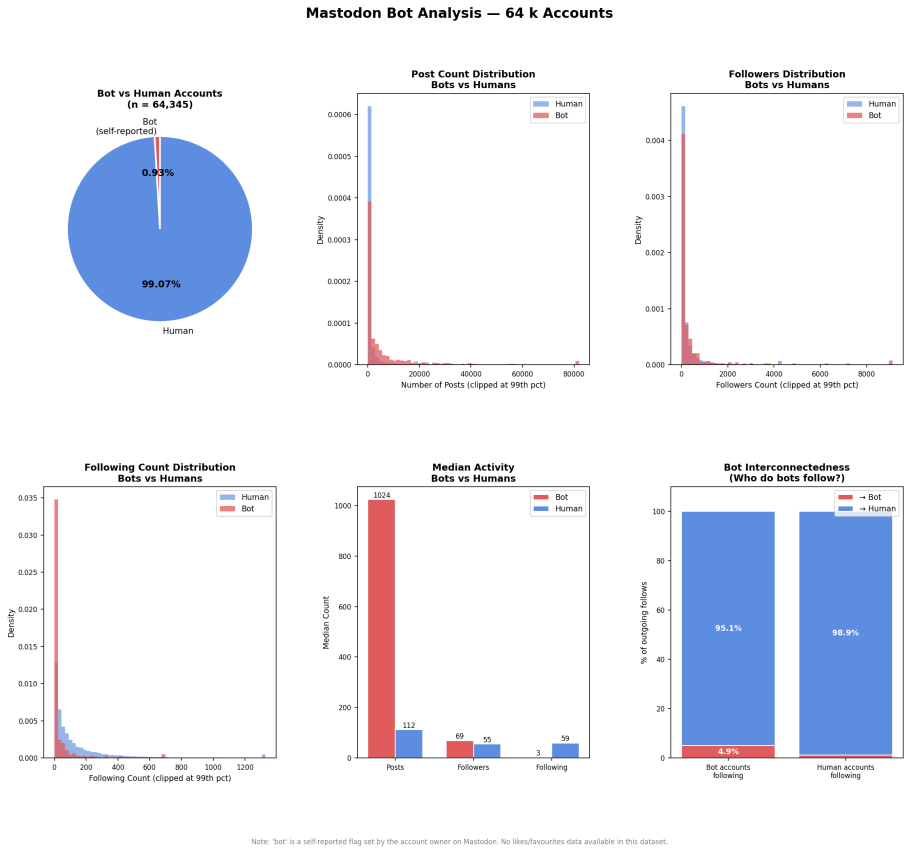


Figure 3.2: Analysis of accounts labeled 'social bot' in FediData and their account activity

As shown in Figure 3.3, ...

3.2 RQ1: Visual Homophily Test

For testing the visual homophily we sampled two groups of users. 10,000 users for each group and one group had always a connection between the two users (connected pair) while the second one didn't (disconnected pair). Additionally, every user for both groups had to have at least 5 images posted on their account so we can meaningfully measure the visual homophily trend. Lastly, we ran a 2 sanity checks. We tested for the trend if every user in the disconnected pair group had at least one other social connection with a different user. It is possible that a user might've been inactive since creation therefore being a lurker in the network. That could then skew the results by artificially increasing the gap of disconnected users. Since we noticed that a particular user had multiple occurances (up to 45 times) second sanity check limited every user in both groups to a single occurance.

Lastly, we were then able to compare the similarity using cosine similarity between the image embeddings obtained earlier. Based on these similarities our Research Question could be answered. However, three different comparison techniques were used to test for the homophily test. Firstly, Mean Pooling: which creates a mean of all embeddings of one user and compares that mean with a mean representing the other user. Second technique

used mean per each pair of images. For every image user A posted we compute similarity for every image of user B. These values are then averaged. Last technique is similar as the second one with the only catch of averaging only the top 3 most similar images of each user in the pair. Which in theory should better represent the depth of similarity of a pair.

For this Research Question to yield any statistically significant results the Mann–Whitney U test was used. This test was used since it makes no assumption about the distribution of two independent groups. Exactly our use case. (TODO: MWU paper to source here)

3.2.1 Results for RQ1 TODO: for now here later move to results with proper discussion

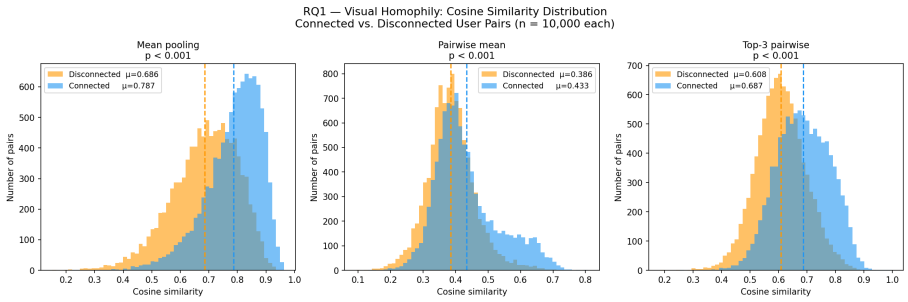


Figure 3.3: Results for RQ1

TODO: comments on graph

Table 3.1: Mann–Whitney U test results for RQ1 visual homophily. For each similarity aggregation method, connected and disconnected pair means (μ) are compared using a one-sided Mann–Whitney U test. Δ is the difference in means. U is the count of pairs out of the 100,000,000 possible combinations where connected pairs were higher

Method	$\mu_{\textbf{conn}}$	$\mu_{\textbf{disc}}$	Δ	U	p
Mean pooling	0.7868	0.6856	+0.1012	75,873,025	< 0.001
Pairwise mean	0.4335	0.3856	+0.0479	63,434,684	< 0.001
Top-3 pairwise	0.6866	0.6081	+0.0786	73,060,093	< 0.001

3.3 RQ2: FAISS Benchmarking

For this research question we

3.4 RQ3: Coordination Detection

Similarity search on disconnected pairs maybe label validation.

3.5 Ethical Considerations

Need to say that FediData is already anonymized by its creators. (All usernames were hashed) I used a public Zenodo release of the dataset, no de-anonymization was done on my end.

Chapter 4

Results

4.1 Overview of Findings

4.2 Results for RQ1

Chapter 5

Discussion

5.1 Interpretation of Findings

Interpret your results in light of your research question and existing literature.

5.2 Comparison with State of the art and current research

Discuss how your findings confirm, extend, or contradict previous studies.

5.3 Limitations and Future Research

Chapter 6

Conclusion and Future Work

6.1 Conclusion